

Multiple Choice (10 marks)

1. The eigenvalues of a Hermitian operator are always
 - (a) real
 - (b) imaginary
 - (c) degenerate
 - (d) linear
2. The emission spectrum of the doubly ionized lithium atom Li^{++} ($Z = 3$, $A = 7$) is identical to that of a hydrogen atom in which all the wavelengths are
 - (a) decreased by a factor of 9
 - (b) decreased by a factor of 49
 - (c) decreased by a factor of 81
 - (d) increased by a factor of 9
3. A three-dimensional harmonic oscillator is in thermal equilibrium with a temperature reservoir at temperature T . The average total energy of the oscillator is
 - (a) $(1/2) k T$
 - (b) kT
 - (c) $(3/2) k T$
 - (d) $3 kT$
4. For which of the following thermodynamic processes is the increase in the internal energy of an ideal gas equal to the heat added to the gas?
 - (a) Constant temperature
 - (b) Constant volume
 - (c) Constant pressure
 - (d) Adiabatic
5. An organ pipe, closed at one end and open at the other, is designed to have a fundamental frequency of C (131 Hz). What is the frequency of the next higher harmonic for this pipe?
 - (a) 44 Hz
 - (b) 196 Hz
 - (c) 262 Hz
 - (d) 393 Hz

True / False (10 marks)

Answer True or False

6. True or False: In a single-electron atom with $l = 2$, there are only 4 allowed values for the magnetic quantum number m_l .
7. True or False: Electromagnetic radiation emitted from a nucleus is most likely to be in the form of gamma rays.
8. True or False: A dye laser is the most suitable type of laser for spectroscopy across a range of visible wavelengths.
9. True or False: A particle that decays in 2.0 ms in its rest frame will travel 360 m in the lab frame at a speed of 0.60 c.
10. True or False: In a reversible thermodynamic process, the entropy of the system and its environment remains unchanged.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the conditions under which a beam of electrons can diffract on a crystal surface, and analyze the relationship between the kinetic energy of the electrons and the observed scattering pattern, considering a lattice spacing of 0.4 nm. What is the approximate kinetic energy required to achieve this diffraction?
12. Discuss the relationship hypothesized by De Broglie between the linear momentum and wavelength of a free massive particle, emphasizing the role of Planck's constant in this context.

End of Paper